

Increasing Participation in AI-Assisted Content Validation Processes: An AI Ergonomics Perspective

Catarina de Brito Carvalho

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Abstract

AI-assisted content validation processes rely on Subject Matter Experts (SMEs) to verify AI-generated outputs before they are incorporated into organizational systems. However, the effectiveness of such human-in-the-loop workflows depends on obtaining sufficient participation from knowledgeable individuals. This review examines how response rates can be increased in AI-assisted content validation processes.

The review synthesizes evidence from survey methodology, social influence, human-AI trust, and human-in-the-loop validation research. Across the literature, communication practices, perceived relevance, trust, and participant effort emerged as more important determinants of participation than the specific characteristics of the feedback interface. The findings suggest that users are more likely to engage when they understand the purpose of the activity, perceive their contribution as valuable, and can complete validation tasks with minimal effort. Transparency regarding the role of AI and the continuing importance of human expertise may further support engagement.

From an AI Ergonomics perspective, participation should be treated as a core design objective of AI-assisted validation systems. Effective validation workflows require balancing response burden with feedback quality while integrating validation activities into existing work practices. The review concludes that successful human-in-the-loop AI systems depend not only on the accuracy of AI-generated outputs but also on the willingness of users to participate in their verification.

1 Introduction

Collecting feedback from users is a fundamental component of evaluating, validating, and improving AI systems. However, obtaining sufficient participation remains challenging. Low response rates can reduce the representativeness and practical value of collected feedback, limiting the effectiveness of systems that rely on human input. AI Ergonomics is concerned with how humans and AI systems can effectively collaborate on shared tasks, emphasizing the design of work processes that support human capabilities, motivation, and well-being while enabling effective human-AI cooperation. From an AI Ergonomics

perspective, participation is not simply a data collection issue but a key factor influencing the success of AI-assisted validation workflows, where the quality and usefulness of the system depend on users' willingness to review and verify AI-generated outputs.

In many human-in-the-loop AI applications, Subject Matter Experts (SMEs) are asked to validate information generated or extracted by AI systems. Examples include reviewing requirements, verifying project responsibilities, confirming knowledge base entries, or assessing automatically generated summaries and classifications. Although advances in AI have reduced the effort required to generate information, human expertise remains necessary to verify its accuracy and reliability.

The motivating use case for this review concerns the validation of AI-generated statements regarding individuals' involvement in project activities. Subject Matter Experts are then asked to confirm, reject, or correct the identified task associations before the information is incorporated into a broader

2 Literature Review

2.1 Survey Methodology and Response Rates

Online surveys remain one of the most widely used methods for collecting feedback and research data due to their scalability, low cost, and ease of deployment. Nevertheless, they continue to face persistent challenges, including lower response rates, non-response bias, measurement errors, and privacy concerns (Zarouali et al. 2024). Consequently, a substantial body of research has focused on identifying practical strategies that increase participation and improve response rates.

Several recurring themes emerge from this literature. First, participation appears to depend heavily on the visibility, accessibility, and perceived legitimacy of the request. Personalized communications, professionally presented invitations, reminders, and the use of multiple contact methods have all been associated with higher response rates (Millar and Dillman 2011; Edwards et al. 2023). Rather than identifying a single optimal intervention, these studies suggest that participation is influenced by the combined effect of several factors, including communication frequency, response flexibility, convenience, and the perceived credibility of the request.

A second theme concerns the perceived value of participation. Surveys addressing topics that participants considered personally or professionally relevant consistently achieved higher response rates (Wu et al. 2022). Similarly, Edwards et al. (2023) found that communications emphasizing societal benefits and meaningful outcomes increased willingness to respond. These findings suggest that participation decisions are influenced not only by the effort required to respond, but also by whether respondents perceive the activity as worthwhile and believe their contribution will have value.

The role of incentives is less straightforward. While Edwards et al. (2023) and Millar and Dillman (2011) reported positive effects of monetary and prepaid incentives, Wu et al. (2022) found no significant overall relationship between incentives and response rates. One possible explanation is that incentives interact with contextual and motivational factors rather than operating as universally effective participation mechanisms. For some respondents, particularly those already invested in a topic, the perceived value of contributing may be more influential than material rewards.

Finally, the literature suggests that questionnaire characteristics themselves may play a smaller role than is often assumed. Porter and Whitcomb (2005) found that questionnaire length was only weakly associated with participation once respondents had already accessed the survey, indicating that many barriers to participation arise before respondents engage with the questionnaire itself. Similar conclusions emerge from research comparing alternative feedback interfaces. Although chatbot-based questionnaires were perceived as more engaging and enjoyable than traditional surveys, these improvements did not consistently translate into higher response rates, improved retention, or better data quality (Zarouali et al. 2024). Taken together, these findings suggest that communication practices, recruitment strategies, and participant motivation are generally more important determinants of participation than the specific characteristics of the feedback mechanism itself.

2.2 Social Influence and Compliance

While survey methodology research identifies practical interventions that increase participation, social influence research helps explain why such interventions are effective. Rather than focusing on survey design, this perspective examines the psychological processes that shape individuals' decisions to engage with participation requests.

One of the most robust findings concerns autonomy. Individuals are more likely to comply with requests when participation is perceived as a voluntary choice rather than an obligation. Consistent with this view, Guéguen and Pascual (2005) found that explicitly emphasizing participants' freedom to accept or decline a request increased survey participation. These findings align with Self-Determination Theory, which identifies autonomy as a fundamental psychological need that supports motivation and engagement.

Participation is also influenced by social identity and perceived relevance. Individuals are more likely to engage when they believe their expertise, experience, or group membership is specifically valued. For example, Dolinski et al. (2024) demonstrated that participation increased when potential respondents were informed that researchers were looking for people like them. The effect appeared to arise from the perception that respondents possessed unique knowledge or experience that was important to the success of the study.

A related mechanism concerns prosocial motivation. Individuals are often more willing to participate when they believe their contribution will help others or contribute to a meaningful objective. This perspective may help explain why messages emphasizing societal benefits frequently increase participation and why altruistic appeals can

sometimes outperform material incentives among highly motivated individuals (Conn et al. 2019). Participation is therefore influenced not only by personal benefit, but also by perceptions of impact and contribution.

Taken together, these findings suggest that participation decisions are shaped not only by practical considerations such as convenience and effort, but also by deeper psychological needs relating to autonomy, identity, and purpose. For AI-assisted content validation processes, framing participation as a meaningful opportunity to contribute expertise and improve the quality of AI-generated outputs may therefore be more effective than relying solely on procedural or technological changes.

2.3 Human-AI Trust

Research on human–AI trust suggests that trust in AI systems is not determined solely by their technical performance. Rather, trust often emerges from comparative evaluations between AI systems and human decision-makers. Galindez-Acosta and Giraldo-Huertas (2026) found that individuals who expressed lower levels of trust in humans were more likely to trust AI systems, suggesting that trust in AI may partly reflect dissatisfaction with human judgement rather than confidence in AI alone.

Users also appear to rely on cognitive heuristics when evaluating AI-supported processes. Molina and Sundar (2024) distinguish between a positive machine heuristic, whereby AI is perceived as objective, accurate, and less biased than humans, and a negative machine heuristic, whereby AI is viewed as rigid, lacking contextual understanding, and incapable of nuanced judgement. These perceptions influence both trust in AI and willingness to rely on AI-generated outputs.

For AI-assisted content validation, such findings suggest that participation may depend not only on the effort required to provide feedback, but also on how users perceive the AI system requesting that feedback. Individuals who view AI as objective and capable may be more willing to engage with AI-generated statements, whereas those who perceive AI as unreliable or lacking contextual understanding may be less inclined to participate. Consequently, response rates may be influenced not only by survey methodology and participation strategies, but also by how the AI system and its role are communicated to potential respondents.

Recent conceptual work further argues that trustworthiness, trust, and trusting behaviour are distinct concepts and that trust in AI is context-dependent, individually variable, and strategically motivated (Everett et al. 2026). From this perspective, the relevant outcome for AI-assisted validation is not simply whether users trust the AI system, but whether they choose to engage with the validation process. Effective validation workflows may therefore require calibrated trust rather than maximal trust, ensuring that users perceive AI-generated outputs as sufficiently credible to warrant engagement while still recognising the need for human verification and oversight (Gupta et al. 2025). Nevertheless, many determinants of trust are shaped by broader organizational and technological factors

beyond the feedback mechanism itself, suggesting that trust may function primarily as a contextual influence rather than a direct intervention target.

2.4 Development and Evaluation of Feedback Mechanisms

Because AI-assisted content validation remains a relatively novel workflow, it may be difficult to predict in advance which feedback mechanisms will achieve acceptable participation rates. Consequently, formative evaluation methods can play an important role in reducing design uncertainty before large-scale deployment.

According to guidance from the Office of Management and Budget (Office of Management and Budget 2006), pretests and pilot studies can identify potential issues in measurement procedures, survey design, participant communication, and operational processes before full implementation. In the context of AI ergonomics, pretests may be particularly useful for evaluating different feedback formats, testing invitation messages, assessing user engagement, and identifying barriers to participation.

In addition to pretests and pilot studies, qualitative methods such as focus groups and cognitive interviews may provide valuable insights during the development of feedback mechanisms. Focus groups can help researchers understand how participants perceive a topic, the language they use to discuss it, and the factors that influence their willingness to engage. In the context of AI-generated content validation, focus groups could be used to explore users' attitudes toward AI, perceptions of feedback requests, and preferences regarding different feedback formats.

Cognitive interviews may be particularly useful when designing feedback questions or validation criteria. By asking participants to verbalize their thought processes while responding, researchers can identify ambiguities, misunderstandings, or unintended interpretations of questions. This approach can help ensure that users interpret validation requests as intended and that the resulting data accurately reflects their evaluations.

2.5 Expert Review and Human-in-the-Loop Validation

Human-in-the-loop (HITL) approaches have emerged as a key paradigm for ensuring the quality, reliability, and accountability of AI-assisted systems. Rather than treating human involvement as a simple alternative to automation, HITL frameworks position humans as active participants who review, validate, correct, or refine AI-generated outputs before they are accepted or acted upon (Lazaros et al. 2025). Human involvement can occur at different stages of the AI lifecycle, including data collection, model training, output evaluation, and decision-making. In the context of AI-assisted content validation, subject matter experts (SMEs) act as validators who assess the accuracy of AI-generated statements and provide feedback that may improve future system performance.

A central theme within the HITL literature is the trade-off between automation and human oversight. Increasing human involvement can improve accuracy, transparency, and accountability, but it also increases the effort required from participants (Lazaros et al.

2025). This trade-off is particularly relevant when designing validation workflows. Asking users to provide detailed corrections, explanations, or supporting evidence may improve the quality of the collected feedback, but may also reduce participation due to the additional time and effort required. Conversely, simpler validation tasks, such as confirming or rejecting AI-generated statements, may encourage greater participation while providing less detailed information. Consequently, the effectiveness of an AI-assisted validation process may depend not only on how participation is encouraged, but also on how validation tasks are structured. Designing workflows that balance response burden with feedback quality may therefore be critical for achieving both high participation rates and meaningful validation outcomes.

Unlike many human-in-the-loop applications where users primarily verify algorithmic outputs, AI-assisted content validation often relies on SMEs possessing contextual or experiential knowledge that is not fully captured within the source documents. Consequently, SMEs act not only as validators but also as providers of ground truth capable of correcting omissions, resolving ambiguities, and contributing information unavailable to the AI system. This further increases the importance of designing validation tasks that encourage participation while minimizing unnecessary burden.

3 Practical Recommendations

The reviewed literature suggests that participation in AI-assisted content validation processes is influenced more by communication, motivation, trust, and validation task design than by the specific feedback technology used. From an AI Ergonomics perspective, participation should therefore be treated as a design objective rather than an operational concern. The following recommendations summarize the principal implications of the reviewed evidence.

Table 1 summarizes the principal recommendations emerging from the literature review.

Recommendations for AI-Assisted Content Validation Processes

Area	Key Recommendations
Development and Testing	Conduct pretests and pilot studies before deployment; use focus groups, cognitive interviews, and think-aloud protocols to identify usability issues, misunderstandings, and barriers to participation. Include both AI-supportive and AI-skeptical participants where possible.
Participation Strategy	Clearly communicate the purpose, value, and expected impact of participation; emphasize the

	importance of participants' expertise; pre-contact participants and use personalized communications whenever possible.
Engagement and Follow-up	Use reminders and familiar communication channels; adopt a multi-channel communication strategy where appropriate (e.g., 1st Webex, 2nd email); target follow-up reminders to non-respondents; provide convenient response mechanisms; consider incentives where appropriate, recognizing that their effectiveness may depend on participant motivation and context.
Validation Task Design	Minimize unnecessary effort; prioritize simple validation actions (e.g., confirm, reject, correct); provide clear instructions and validation criteria; prioritize relevance and clarity over simply reducing questionnaire length.
Human-AI Collaboration	Explain how AI-generated outputs were produced; acknowledge system limitations; emphasize the continuing importance of human expertise; present validation as a collaborative human-AI process.
Evaluation and Improvement	Monitor response rates, completion rates, feedback quality, user satisfaction, perceived effort, and long-term participation; continuously refine communication strategies and validation workflows based on collected data.

Conclusion

This review examined the question of how response rates can be increased in AI-assisted content validation processes. Across the reviewed literature, communication practices, perceived relevance, social influence, trust, and validation task design emerged as more

important determinants of participation than the specific characteristics of the feedback technology itself. Participation was found to be influenced not only by practical considerations such as convenience and effort, but also by psychological factors including autonomy, perceived contribution, and confidence in the validation process.

From an AI Ergonomics perspective, these findings suggest that participation should be treated as a central design objective of human-in-the-loop AI systems. The effectiveness of AI-assisted validation depends not only on the quality of AI-generated outputs, but also on the willingness of knowledgeable individuals to engage in their verification. Consequently, successful validation workflows require careful attention to communication, transparency, motivation, and workload, ensuring that human expertise can be integrated into the system in an efficient and sustainable manner.

Ultimately, the challenge of AI-assisted content validation is not simply generating information that can be reviewed by humans, but designing processes that encourage humans to participate in that review. From an AI Ergonomics perspective, participation should be regarded as a design objective rather than an operational afterthought, requiring careful consideration of motivation, workload, transparency, and the integration of validation activities into existing work practices.

Future research should evaluate these recommendations empirically within real-world AI-assisted validation workflows to determine which combinations of interventions produce the greatest improvements in participation and feedback quality. As organizations increasingly rely on human-in-the-loop approaches to validate, refine, and govern AI-generated outputs, participation may become as critical to system success as the performance of the underlying AI models themselves.

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